Aryan Mondal

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Education

University of Massachusetts Amherst — Amherst, MA
MS in Computer Science — *Sep 2026 – Present*

- **Relevant Coursework**: COMPSCI 660 Advanced Information Assurance, COMPSCI 590AF Reverse Engineering & Understanding Exploit Development, COMPSCI 589 Machine Learning

SRM Institute of Science and Technology — Chennai, Tamil Nadu
B.Tech in Computer Science and Engineering, CGPA: 8.39/10 — *Sep 2022 – May 2026*

- **Relevant Coursework**: Cryptography and Network Security, Forensics and Incident Response, Security Management, Computer Networks, Cyberwarfare, Mobile and Wireless Security, Hacker Techniques, Tools, and Incident Handling

Experience

Security Intern — Feb 2025 – Mar 2025
Thirumoolar IT Solutions — *Chennai, Tamil Nadu*

- Implemented 20+ vulnerability scans and penetration tests using OWASP ZAP, Burp Suite, and Nmap, identifying 40+ security issues and contributing to a 25% improvement in system security posture.
- Investigated codebases of 3 internal web applications by applying manual code analysis, detecting SQL Injection and XSS vulnerabilities and enabling developers to resolve a significant share of exploitable security flaws.

Publications

An Adaptive Feedback Architecture for Stabilizing SNSPDs in TF-QKD Networks using SDQN — Under Review
IEEE Transactions on Quantum Engineering (TQE)

- Engineered an SDQN with hot-swappable PID and Bayesian optimization controllers by decoupling the classical control plane from the quantum data plane, achieving a 41% Secret Key Rate improvement and 4× reduction in SKR variance (σ_{SKR}: 0.180 → 0.038).
- Modeled SNSPD photon detection efficiency using Ginzburg–Landau critical current equations by deploying a Matérn 5/2 Gaussian Process with Expected Improvement acquisition, converging to 95% of the optimal operating point in 80 cycles versus 120 for PID alone.
- Implemented Mahalanobis Distance anomaly detection across correlated SKR, QBER, temperature, and bias metrics by triggering an automated Security Mode clamp upon a 3σ breach, defeating blinding and thermal manipulation attacks.

A Robust and Lightweight Authentication Protocol for UAV Swarm Communications using HKDF-based Key Derivation and AEAD — Jul 2026
IEEE CONECCT 2026

- Designed R-HAAP by replacing a Duffing chaotic-map keystream with a verified ECDH → HKDF → ChaCha20-Poly1305 pipeline, achieving 150–250 μs per-packet latency on ARM Cortex-M4 — 20–33× faster than the original protocol and immune to cache-timing side-channel attacks.
- Implemented per-packet key derivation by constructing a unique HKDF.expand info string from UAV ID and monotonic offset counter, cryptographically binding metadata to the Poly1305 MAC and providing provable replay and forgery resistance.
- Validated swarm scalability using UavNetSim-v1 by simulating star and FANET mesh topologies across 10–100 UAVs, demonstrating 99.8%+ packet delivery and under 15 ms end-to-end latency within the 100 ms real-time flight control threshold.

Lightweight Hybrid DNN-GNN Architecture for Network Intrusion Detection with Adaptive Late Fusion — Apr 2026
ICNWC 2026

- Designed a hybrid DNN (∼68K params) and Pseudo-GNN (∼8K params) with adaptive late fusion by combining parallel statistical and relational branches, achieving 99.64% accuracy on 60K CIC-IDS-2017 samples at 1,250× fewer parameters than Transformer-based NIDS.
- Analyzed a learnable fusion weight α by tracking its convergence from 0.689 to 0.389 over 25 epochs, establishing relational features as more discriminative and confirming the 8K pseudo-GNN branch outperforms the 68K DNN-only baseline.
- Benchmarked the model against ConvNeXt-Tiny (28M params) and a Transformer NIDS (100M params) by evaluating on 9,000 held-out samples, achieving competitive accuracy at 350× and 1,250× fewer parameters with a 0.08% false positive rate.

Quantum Key Distribution for SCADA in Power Grids: Security Architectures and Practical Applications — Nov 2025
ICETCE 2025

- Developed SVM and Random Forest classifiers for QKD-enhanced SCADA anomaly detection by training on 10,000 simulated data points via GridSearchCV, achieving 94.9% accuracy with SVM attaining 100% recall.
- Evaluated four classifiers — SVM, Random Forest, KNN, and Neural Network — by comparing precision (0.68–0.70), recall (0.94–1.00), and F1-score (0.80–0.81), identifying SVM as optimal and exposing KNN's complete failure under class imbalance.
- Established a replicable QKD-SCADA security framework by researching 30+ field deployments across Tokyo, Vienna, and China's UHV networks, mapping BB84 and decoy-state QKD architectures to SCADA threat vectors including FDIA and MitM.

Technical Skills

Programming Languages: Python, C, C++, Bash, SQL
Cyber Security & Networking: Cryptography, Digital Forensics, Anomaly Detection, Malware Analysis, Penetration Testing, Web Exploits SQLi/XSS, Operating Systems, SIEM/SOC Operations, SCADA Security, UAV Security, Quantum Key Distribution QKD
Tools & Frameworks: Claude Code, OWASP ZAP, Burp Suite, Nmap, Wireshark, Scapy, PyShark, Apache Kafka, FastAPI, Grafana, TensorFlow, Keras
Other Skills: Risk Management, Compliance & Regulation, Security Awareness, Threat Intelligence, Strategic Planning, Project Management